

Chapter 2

Role of Government in the Economy

Public finance is concerned with the role of government in the economy. The field addresses questions such as the following:

- Why and how do governments intervene in the economy?
- What are the economic effects of government intervention?

Before the next sections cover the answers to those question, we cover the difference between positive and normative economics. Positive economics is the objective analysis of economic outcomes (e.g., the effect of increasing interest rates on the number of mortgage applications). Normative economics includes value judgements of what should be done (e.g., evaluation of various policies to increase education spending to make society better off). Positive economics can inform normative economics. Of course, the boundaries between both concepts are not always clear cut because even positive economics may include some value judgement given the assumptions made. In public finance and public management economics, we often need to make value judgements in an attempt to make society better off.

Below, we are covering three reasons why governments intervene in the economy:

1. Market failures
2. Economic stabilization
3. Redistribution

2.1 Market Failures

The First Theorem of Welfare Economics can be considered the theoretical basis for the intervention of government in the economy:

If (1) households and firms act perfectly competitively, taking prices as given, (2) there is a full set of markets, and (3) there is perfect information, then a competitive equilibrium is efficient.

The implications of the First Welfare Theorem are that Pareto efficiency can be achieved in a competitive market. A competitive market is usually the benchmark to judge actual market outcomes. The First Welfare Theorem requires the absence of market imperfections such as (1) public goods, (2) externalities, (3) imperfect competition, and (4) asymmetric information. Note that the term “public” refers to a characteristic of a good and not the provider.

Some characteristics of a perfectly competitive market are that (1) there are many producers and many consumers, (2) free entry and exit into the market exists, and (3) market participants are engaging in price-taking behavior. The price-taking behavior is a direct result of many producers and many consumers. That is, each market participant is aware that no matter what quantity of the product they supply or demand at the individual level, the price does not change. Market imperfections and failures exist and lead to an inefficient allocation of societal resources.

2.1.1 Public Goods

As mentioned before, “public” is a reference to characteristics and not the provider of the good. The distinction between various types of goods (e.g., public good versus private good) can be conducted along two dimensions:

- *Rivalry*: Overall consumption of the good is affected by individual consumption.
- *Excludability*: Ability (or lack thereof) to exclude people from consuming the good.

Consider the following table to distinguish four types of goods along those two dimensions.

	Rival	Non-Rival
Excludable	Private good	Club good
Non-Excludable	Common good	Public good

Here are some examples of the various types of goods

- *Private goods*: Sandwiches, laptops, etc.
- *Club goods* (also known as *toll goods*): Movies in theaters, toll roads, concerts, etc.
- *Common goods* (also known as *common pool resources*): Aquifers, fishing grounds, petroleum reserves, etc.
- *Public goods*: National defense, mosquito abatement, pollution control, disease control, lighthouses, GPS, etc.

Club goods and public goods are under-produced and common goods are over-consumed. The marginal (additional) cost to provide a public good to one more person is zero.

2.1.2 Externalities

An externality is a positive or negative effect resulting from a person's (or entity's) action without them receiving any compensation or incurring any cost. The distinction is made between a positive and negative externality:

- *Positive externality*: Production of uncompensated benefits received by others (e.g., landscaping, beekeeping, university, research and development). Goods that generate a positive externality are under-produced.
- *Negative externality*: Production of uncompensated cost borne by others (e.g., Gulf of Mexico Hypoxic Zone, cigarettes). Goods that generate a negative externality are over-produced.

We can examine the history of firefighting to see how a negative externality led to the development of public fire services. A century back, homes purchased (or not) fire insurance from companies. The house was then marked with a mark showing the insurance company. If a house was on fire, the fire services sent had to match the insurance company. That is, if your uninsured neighbor's house was burning, fire services would not arrive for them.

There are also pecuniary externalities, which are the result of actions by market participants having an effect on market prices. For example, a sudden increase in home purchases increases the market price making it more expensive for other consumers. But it is also more profitable for homeowners selling and under complete markets, there is a net zero effect. Hence, pecuniary externalities are not considered (from an economic perspective) problematic.

2.1.3 Imperfect Competition

The requirements of the First Welfare Theorem regarding perfect competition are many producers and many consumers, free entry and exit into the market, and price-taking behavior. Market structures for which those conditions are violated suffer from an efficiency loss. Those market structures are either a monopoly (one producer) or oligopoly (small number of producers). Market inefficiencies also exist when the number of consumers is small or singular. In that case, the terms oligopsony and monopsony are used. Underproduction of goods occurs in the context of imperfect competition.

2.1.4 Other Market Failures

Two other types of market failures are asymmetric information and adverse selection. Asymmetric information refers to a situation in which one party in an interaction (e.g., contract, transaction) has more or better information than the

other party. The most common example is the purchase of a used car. In this case, the seller has more information than the buyer. The buyer having more information of the seller can also occur for example in the case of obtaining health insurance. Adverse selection leads to market participation based on asymmetric information (e.g., skydivers purchasing more insurance, bad drivers choosing a lower deductible).

Moral hazard refers to a change in risk engagement due to presence of insurance. For example, the presence of (subsidized) flood insurance may decrease the hesitation to move to a flood prone area. The premium for federal flood insurance is about 38% of what it should be in a competitive market.

Two following motivations of governments to intervene in the economy are covered in the next sections:

- **Economic stabilization:** Insuring economic growth as well as preventing high unemployment and inflation
- **Redistribution:** Transfer payments to ensure a minimum standard of living for all members of society

2.2 Economic Stabilization

A critical function of the government is to stabilize the economy and ensure a low and stable inflation rate as well as low unemployment rate. There are two types of interventions:

- *Monetary policy:* Monetary policy such as setting the interest rate is conducted by the central bank.
- *Fiscal policy:* Fiscal policy—which is one aspect of this book—includes determining taxes and expenditures. It includes the redistribution of income through transfer payments from one entity to another, such as social security or unemployment benefits. Fiscal policy also involves levying taxes on income, corporate profits, and specific items like tobacco and alcohol. These so-called “sin” taxes are designed to alter consumption patterns and reduce the consumption of products resulting in negative externalities.

No matter what policy intervention is chosen, there is usually a time lag between implementation and policy effects.

2.3 Redistribution

The General Social Survey (GSS) in 2022 asked the following question:

We are faced with many problems in this country, none of which can be solved easily or inexpensively. I'm going to name some of these problems, and for each one I'd like you to name some of these

problems, and for each one I'd like you to tell me whether you think we're spending too much money on it, too little money, or about the right amount. Are we spending too much, too little, or about the right amount on welfare?

Respondents were split on the question with answers of "too little" (34.2%), "about right" (34.7%), and "too much" (31.1%). However the range of answers indicate that some redistribution is desired.

In a market economy, income is distributed to individuals based on their ownership of valuable resources, such as property, skills, and land. However, this distribution is often unequal, as not everyone has equal access to these resources. This creates a motivation for redistribution, driven by both social conscience and the fear of social disorder. Societies may feel a moral obligation to support those with fewer resources, and there is also a practical concern that extreme inequality could lead to instability.

To address these issues, governments implement income redistribution policies to improve the standard of living for low-income individuals and households. This is typically achieved through progressive taxes, where those with higher incomes pay a larger percentage of their earnings, and through transfer programs that provide financial assistance to those in need. These measures aim to create a more equitable society by reducing the disparities in income and wealth.

2.4 Methods of Government Intervention

As mentioned before, monetary and fiscal policy are the broad categories of how governments can intervene in the economy. Governments also play a significant role in the provision and production of goods and services, particularly through infrastructure investments that require funding from taxes. These projects are essential for public welfare but often necessitate substantial financial resources, leading to the collection of taxes to support government initiatives. Government influence extends to private production as well. This influence is exerted through regulation, as well as through subsidization or taxation of private production to guide economic behavior. Additionally, governments purchase goods and services from the private sector, further impacting the market.

We are going to differentiate between provision and production of goods and services. Provision refers to the responsibility of ensuring that a service is available to the public, often involving funding or regulation by the government. Production refers to the actual creation or delivery of that service, which can be done by either the government itself or by private entities.

The production and provision of goods and services is undertaken by three main sectors: private, public, and nonprofit. A central question arises regarding who will provide certain goods and services, particularly those whose societal benefits exceed the costs of production but might not be provided by the market.

This leads to considerations about whether these goods and services should be produced by the public or private sector.

Public provision and public production are seen in entities such as local school districts, state police departments, the Federal Emergency Management Administration (FEMA), and the Bureau of Motor Vehicles. These services are funded and operated by government agencies to meet public needs. In some cases, there is private provision but public production. For example, private entities may provide services like police control at rock concerts, commercial satellites launched by NASA, and fee-based services at state parks. Here, private organizations offer the service, but the government handles the production aspect. Conversely, there is public provision but private production, such as with private prisons, charter and voucher schools, private military contractors, and most weapon systems. In these scenarios, the government funds or mandates the service, but private companies are responsible for production. Finally, there is private provision and private production, exemplified by private schools and the Indiana Toll Road. In these cases, both the provision and production of goods and services are handled entirely by the private sector, often driven by market demand rather than public funding or government mandates. Examples in this category are Indiana toll roads.

An important question relates to the economic effects of government intervention. Economic theory shows that efficiency can be (re-)established after correcting market failures. An example would be a Pigouvian tax for a negative externality or a price ceiling for a monopoly. There is also the question of equity versus efficiency, i.e., protection of certain groups may outweigh efficiency from a political perspective. Of course, there is always the issue of interaction between policies. For example, there is a federal subsidy on electric vehicles but state registration fees on electric vehicles.

2.5 Exercises

1. **Government Intervention** (*): Consider the following government policies: (1) A tax on carbon emissions, (2) government funding for national defense, and (3) food assistance for low-income households. For each policy:
 - a. Identify the main economic rationale for government intervention.
 - b. Explain the underlying problem that may prevent an unregulated market from producing a desirable outcome.
 - c. Explain how the government policy is intended to address that problem.
 - d. State one potential limitation or unintended consequence of the intervention.
2. **Businesses and Education** (**): Private businesses benefit from a well-educated workforce because today's students are potential future employees. Yet businesses generally contribute relatively little directly to primary and

secondary education beyond the taxes they pay. Use a generative AI tool to answer the following question: “Is primary and secondary education a public good, and if businesses benefit from education, why do they not voluntarily pay more to support it?” Then answer the following:

- a. Evaluate the AI’s classification of primary and secondary education using the concepts of rivalry and excludability. Is education a pure public good? Explain.
 - b. Identify at least one positive externality from education that benefits private businesses.
 - c. Explain why an individual business may choose not to contribute voluntarily to education even when the combined benefits to local businesses exceed the cost of providing additional education.
 - d. Identify one claim or piece of reasoning in the AI response that you think is incomplete, misleading, or especially important. Explain your assessment using concepts from this chapter.
3. **Smoking** (*): One rationale for imposing taxes on cigarette consumption is that smokers impose negative externalities on the rest of society for example through secondhand smoking. In the absence of taxation, will people tend to consume too much, too little, or the right amount of cigarettes?
4. **Fishing** (**): There are various indications that the fish population in some fisheries is declining (see for example [Are cod ‘severely depleted’ in the Gulf of Maine? Why fishermen, scientists view ocean depths differently](#)). To stop the depletion, fishing bans have been imposed in various regions globally ([Frank et al., 2005](#)). Suppose a fishery contains a stock of fish that can support a sustainable annual harvest of 1,000 metric tons. There are 20 fishing firms operating in the fishery.
 - a. Using rivalry and excludability, classify the fishery as a private good, public good, club good, or common-pool resource. Explain both dimensions of your classification.
 - b. Use two specific observations or statements from the assigned article to explain how the economic characteristics identified in part (a) appear in an actual fishery. Cite the relevant portions of the article.
 - c. Consider three possible policies: (1) A limit on the total quantity of fish that may be caught by all fishermen or (2) a total allowable catch of 1,000 metric tons allocated through nontransferable quotas (3) or a total allowable catch of 1,000 metric tons allocated through transferable fishing rights. For each policy, explain how it changes the incentives facing an individual fishing firm. Do not simply describe how the policy works.
5. **Agency Budget I** (***) : This exercise involves learning to read and use information in public budgets. As part of this assignment you will be looking at a budget of an U.S. federal agency. First, pick an agency (do not pick a state) from [A-Z index of U.S. government departments and agencies](#)

and make sure that the agency publishes a budget online. Download the budgeting information of the agency. Review the agency's budget and answer the questions below. Note that every budget will not provide sufficient information to answer all of these questions. If the information is not available, you should note this in your answers.

- a. What is the mission of this agency?
- b. What are the major programs carried out by this agency?
- c. What is the percent increase in total requested expenditures for the next year over the last year's budget?
- d. What object of expenditure category is the largest? Which is growing the fastest?
- e. Is the number of personnel (full-time equivalent also known as FTE) in the agency growing, and is the growth rate faster or slower than the personnel budget (in dollars)? What does this tell you about changes in average salaries?
- f. What information does this budget give you on the productivity and performance of this agency?
- g. What information would you like to have to be able to assess the 'efficiency' and 'effectiveness' of this agency?

6 Rodent Control ():** A city is considering whether to subsidize private rodent-control activities. Let Q denote the number of properties receiving rodent-control treatment. The marginal private benefit of treatment is $MB = 60 - Q$ while each additional treated property also reduces rodent problems on neighboring properties. The marginal external benefit is $MEB = 20$. The marginal cost of treatment is constant at $MC = 20$.

- a. Without government intervention, determine the number of properties that receive treatment, (Q_m). Show the condition you use to determine the market outcome.
- b. Construct the marginal social benefit function and determine the socially optimal number of treated properties Q^* .
- c. Calculate the total external benefit generated when treatment increases from Q_m to Q^* . Illustrate this area on a graph showing MB , MSB , and MC .
- d. The city proposes a per-property subsidy to induce households to choose Q^* . Determine the minimum subsidy necessary and calculate the total government expenditure on the subsidy if exactly Q^* properties participate.
- e. A city council member argues: "*The subsidy should equal the full \$20 cost of rodent treatment because otherwise households will still have to pay something themselves and will therefore underinvest.*" Evaluate this argument using your results above.
- f. Now suppose the external benefit is not constant. Instead, $MEB = 30 - 0.5 \cdot Q$. Recalculate Q^* . What should the per-unit Pigouvian subsidy be at the new social optimum? Explain why the optimal subsidy is no longer simply the same amount for every possible level of rodent control.
- g. The city has enough money to subsidize only 10 additional treatments beyond the unregulated market outcome. It must choose between: (1) Providing the same subsidy to every participating household or (2) targeting the subsidy to neighborhoods with the highest estimated spillover benefits. Explain what additional information the city would need to determine which policy produces the larger increase

in social welfare. h. Based on your analysis, explain why observing too little private spending on rodent control does not, by itself, establish that government intervention will improve welfare. Identify at least two pieces of information policymakers would need before reaching that conclusion.

7 Public Goods and Congestion (*)**: Note that for this question, a [template spreadsheet](#) is available. A city of 100,000 residents considers building a public park. Each resident's utility from the park is

$$U_i = v \cdot \ln(G) - \phi \cdot \frac{N}{G} - T$$

where T is the tax per person, G is the park size, N is the number of users, $v = 80$, and $\phi = 30$. The cost is $C(G) = 200G + 0.05G^2$.

1. Calculate the socially optimal G^{***} by maximizing total surplus. I strongly encourage the use of the add-on Solver in Excel.
2. If financed with a head tax, how much would each resident pay, and what would their net utility be?
3. Suppose the city instead leaves funding to voluntary contributions. Explain (and calculate if possible) why the outcome differs.